

ChromaIntel LC-MS/MS MVP Dashboard

Dataset size, model comparison, parameter significance, and development roadmap

4978

Master rows

4978

Model rows

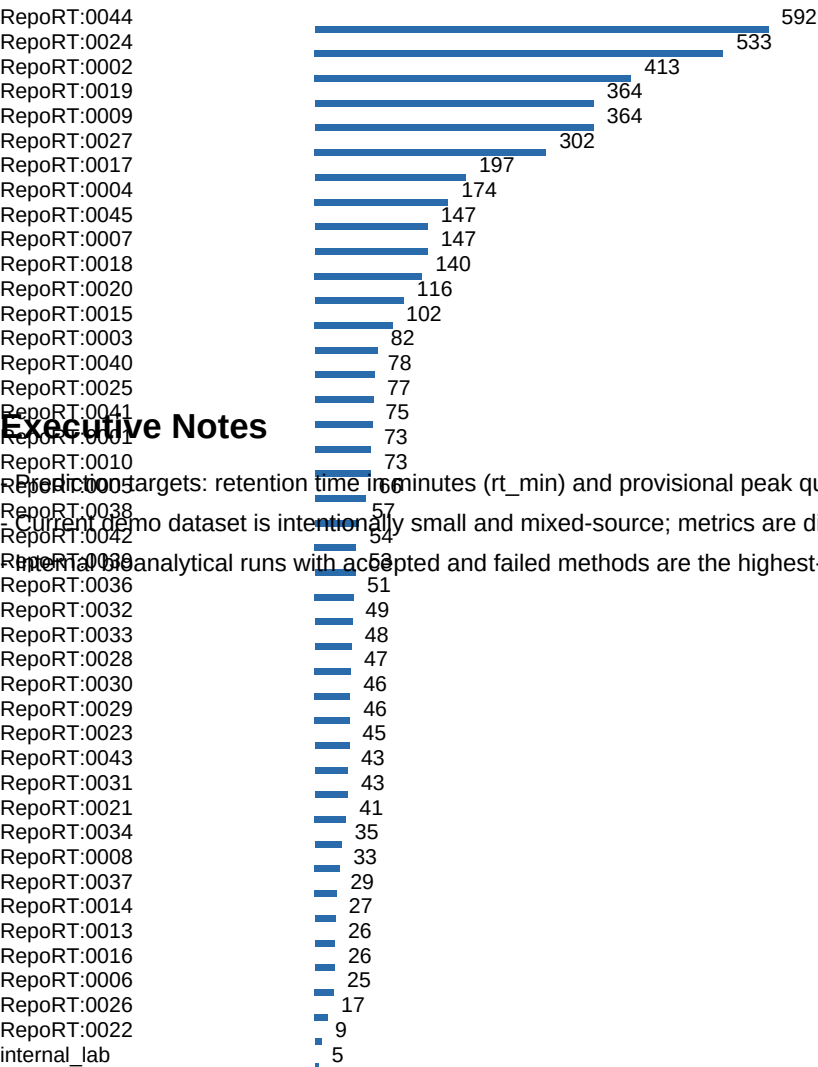
3681

Compounds

46

Sources

Rows by source



Executive Notes

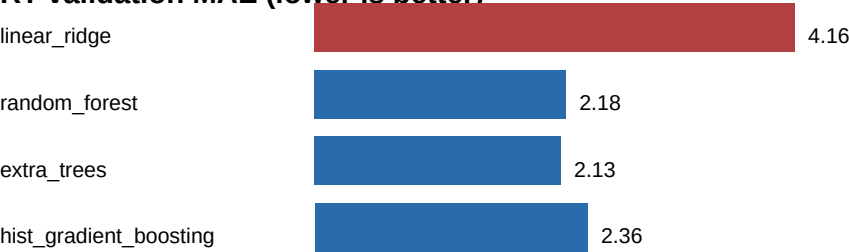
Production targets: retention time in minutes (rt_min) and provisional peak quality score (quality_score).
Current demo dataset is intentionally small and mixed-source; metrics are diagnostic, not production evidence.
Internal analytical runs with accepted and failed methods are the highest-value next data source.

Highest missingness fraction

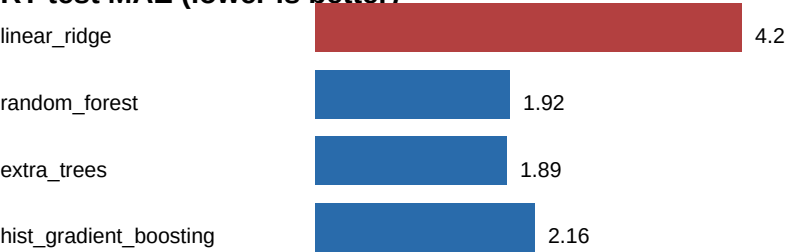


Model Comparison

RT validation MAE (lower is better)



RT test MAE (lower is better)



RT model	Val MAE	Test MAE	Test R2
linear_ridge	4.156	4.202	0.578
random_forest	2.176	1.918	0.881
extra_trees	2.130	1.887	0.864
hist_gradient_boosting	2.363	2.162	0.852

Quality model	Val MAE	Test MAE	Test R2
linear_ridge	0.000	0.000	0.806
random_forest	0.000	0.000	0.999
extra_trees	0.000	0.000	0.380
hist_gradient_boosting	0.000	0.001	0.416

Selected RT model: extra_trees; selected quality model: random_forest

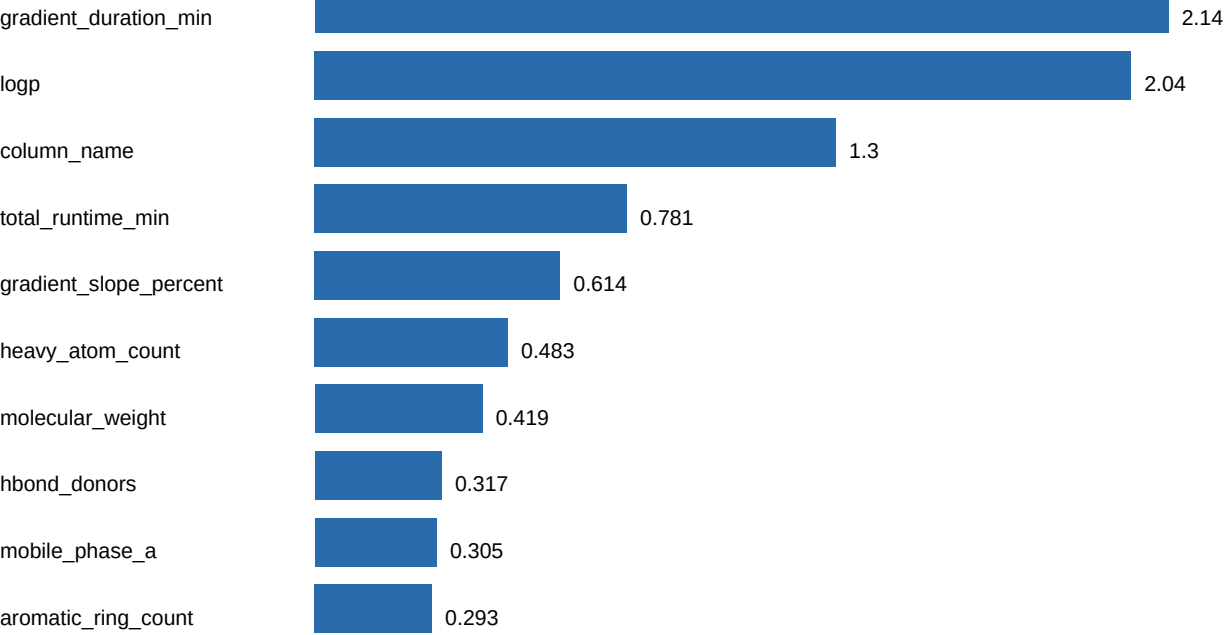
Uncertainty proxy: split_conformal_abs_residual_q90 q90=4.999 min

Source-wise RT error

Source	N	MAE	Bias
RepoRT:0039	9	18.655	12.929
RepoRT:0022	1	8.176	8.176
RepoRT:0037	5	7.602	-6.452
RepoRT:0032	11	6.271	4.678
RepoRT:0007	21	3.945	-0.054
RepoRT:0036	7	3.703	0.653
RepoRT:0006	7	3.611	3.464
RepoRT:0028	13	3.434	-0.448
RepoRT:0002	88	3.363	-0.285
RepoRT:0017	38	3.210	0.557
RepoRT:0021	11	2.934	0.408
RepoRT:0040	14	2.695	-0.574

Parameter Significance And Validation Detail

Top RT permutation importance



Feature	Group	Imp.	Signal
gradient_duration_mi	lc_numeric	2.136	positive
logp	compound_descript	2.042	positive
column_name	lc_categorical	1.304	positive
total_runtime_min	lc_numeric	0.781	positive
gradient_slope_perce	lc_numeric	0.614	positive
heavy_atom_count	compound_descript	0.483	positive
molecular_weight	compound_descript	0.419	positive
hbond_donors	compound_descript	0.317	positive
mobile_phase_a	lc_categorical	0.305	positive
aromatic_ring_count	compound_descript	0.293	positive

Held-out prediction diagnostics

Test rows: 996 | RT MAE: 1.887 min | RT RMSE: 3.869 min | AD flags: 0

Compound	Source	Actual RT	Pred RT	Abs Error
Protocatechuic acid	RepoRT:0039	29.730	72.947	43.217
betulinic acid	RepoRT:0002	53.200	13.388	39.812
malic acid	RepoRT:0039	8.060	38.070	30.010
Epicatechin gallate	RepoRT:0039	58.600	85.489	26.889
Ellagic acid	RepoRT:0032	14.200	38.914	24.714
chinchonidine sulphate	RepoRT:0002	19.150	42.110	22.960
Tartaric Acid	RepoRT:0039	7.510	28.176	20.666
p-Coumaric acid	RepoRT:0039	56.180	74.380	18.200
Ethyl Caffeate	RepoRT:0039	92.700	75.496	17.204
Curcumin	RepoRT:0028	37.720	22.556	15.164
methysalicylate ester	RepoRT:0002	34.500	19.786	14.714

Roadmap

0-2 weeks: data foundation

- Import internal BE-style historical runs with accepted, failed, low-intensity, poor-resolution, and carryover cases.
- Expand reviewed public RT samples and preserve source/license/missingness provenance.
- Add import preview summaries for invalid rows, duplicate runs, and missing MS/LC fields.

2-4 weeks: stronger validation and models

- Move from random split to grouped/source holdout by InChIKey, source, and method family.
- Evaluate Morgan fingerprints, ExtraTrees, CatBoost/LightGBM/XGBoost when enough rows exist.
- Persist applicability-domain ranges and penalize out-of-domain recommendations.

1-2 months: recommendation quality

- Add score decomposition: RT fit, quality, runtime, confidence, AD penalty.
- Make LC search space configurable by lab-approved columns, solvents, pH, flow, temperature, and runtime.
- Collect outcome feedback from recommended methods for active learning.

2-4 months: SOTA research track

- Pretrain or benchmark graph/Transformer structure encoders on METLIN/RepoRT/MCMRT-scale RT data.
- Use retention-order objectives where absolute RT does not transfer cleanly between methods.
- Add calibrated conformal intervals by matrix, source, column chemistry, and instrument platform.

Decision gate: do not claim production readiness until internal data volume, grouped validation, and lab acceptance criteria are in place.